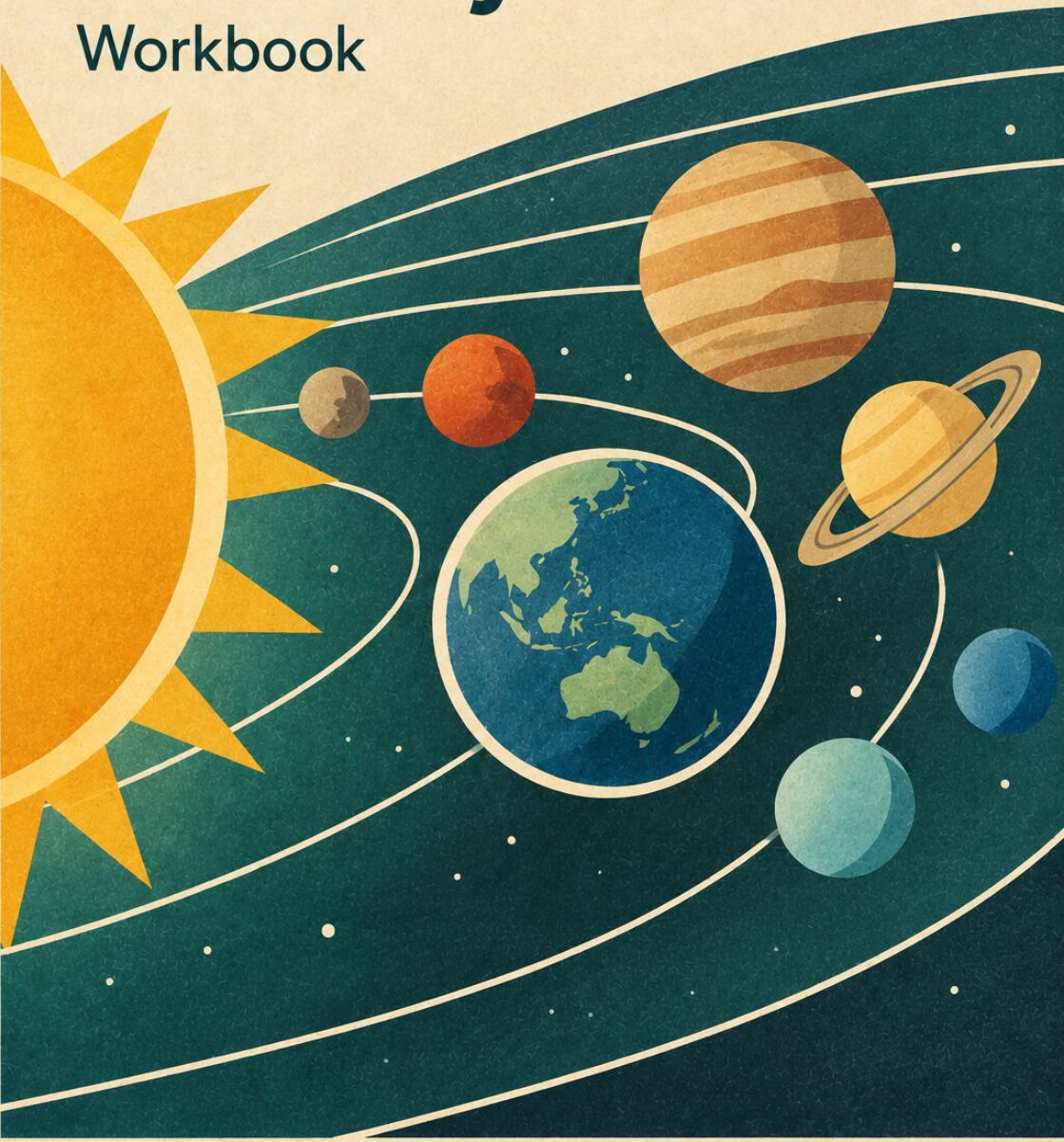
Year 7 Science

Earth in our Solar System

Workbook



Space

Section 1 — A5 trial of the workbook layout

Name _____

Class _____

Teacher _____

◎ Acknowledgement of Country

This booklet was created on the unceded Country of the Wurundjeri-Woiwurrung people of the Kulin Nation. We acknowledge Elders past, present, and emerging. Aboriginal and Torres Strait Islander peoples have read, recorded and navigated by this sky for tens of thousands of years — the oldest continuous astronomical tradition on Earth — and that knowledge appears throughout this unit as living science, not history.

◆ How this booklet works

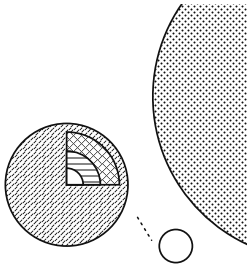
Section N goes with Lesson N. Each kind of box has its own frame and mark:

I do •	Your teacher works it through — you follow
We do ••	You try it with help, together
You do •••	On your own: check your understanding
Key idea	The idea to remember
Above level	Year 8 extension (double frame)
Country	First Nations perspectives (dotted frame)

SPACE • LESSON

1

Earth, Moon and Sun — Scale and Structure



VC 2.0: Science Understanding — Physical sciences: Earth and space sciences; scale and relative size of Earth, Moon and Sun

Key vocabulary

Write what each word means as you meet it in this lesson.

Word	What it means
Empty space	
Inner core	
Outer core	
Mantle	
Crust	

Word	What it means
Satellite	
Regolith	
Tidal locking	
Highlands	
Maria	
Main sequence star	
Nuclear fusion	

1.1 The scale of Earth, Moon and Sun

The three most important objects in our immediate cosmic environment differ enormously in size. Understanding their relative sizes is foundational to everything else in this unit.

Object	Diameter (km)	Mass (kg)	Distance from Earth
Moon	3,474	7.34×10^{22}	384,400 km avg
Earth	12,742	5.97×10^{24}	—
Sun	1,392,000	1.99×10^{30}	150,000,000 km (1 AU)

◆ Key concept: Scale

If Earth were the size of a tennis ball (6.7 cm across), the Moon would be a marble (1.8 cm) about 1.9 m away, and the Sun would be a sphere 7.5 m across, placed 807 m down the road. The solar system is almost entirely **empty space**.

••• You do • Check your understanding

Q1.1
Use the table to state the diameter of Earth and the diameter of the Sun.

Q1.2
Calculate how many times wider the Sun is than Earth. Show your working and give your answer to the nearest whole number.

Q1.3
The Moon is 384,400 km from Earth. Calculate how many Earth diameters would fit into that gap.

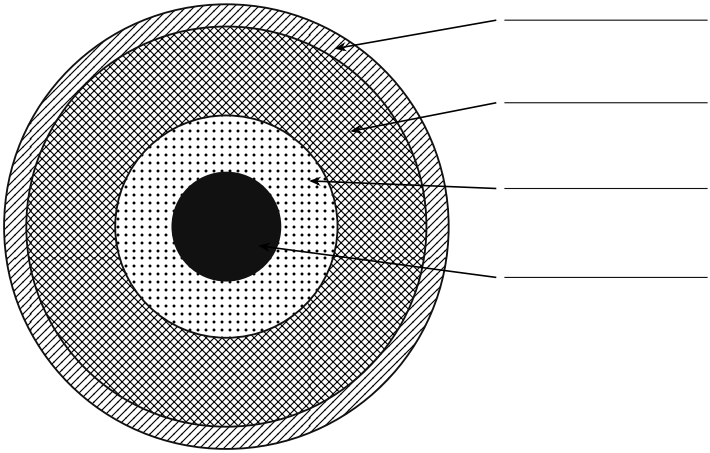
Q1.4
In the tennis-ball model, Earth is 6.7 cm across and the Sun sits 807 m down the road. Explain what this model shows that the table of diameters alone does not.

1.2 Structure of Earth

Earth has four main layers:

- **Inner core** — solid iron-nickel; radius ~1,200 km; temperature ~5,400°C
- **Outer core** — liquid iron-nickel; generates Earth's magnetic field (the dynamo effect)
- **Mantle** — semi-solid rock; 2,900 km thick; convection currents drive plate tectonics
- **Crust** — thin, solid rock; 5–70 km thick; oceanic (denser, thinner) and continental (less dense, thicker)

Diagram: Label the layers of Earth



Word bank: Outer core • Crust • Inner core • Mantle

••• You do • Check your understanding

Q1.5
Which layer generates Earth's magnetic field, and what is this process called?

Q1.6
The mantle is described as semi-solid rock. State its thickness and what its convection currents drive.

Q1.7
Compare oceanic and continental crust using two differences given in the booklet.

1.3 The Moon

The Moon is Earth's only natural **satellite**. It produces no light of its own — everything we see is reflected sunlight.

- **Diameter**: 3,474 km — about one quarter of Earth's
- **Distance**: 384,400 km from Earth on average; light crosses this in **1.28 seconds**
- **Orbital period**: 27.3 days — and it *rotates* in exactly the same time
- **Surface**: covered in **regolith**, a fine powdery rock produced by billions of years of meteorite impacts
- **No atmosphere** and no liquid water, so temperatures swing from -173 °C at night to $+127$ °C in daylight

◆ Why we only ever see one side

The Moon's rotation period and orbital period are identical (27.3 days), so the same face is always turned towards Earth. This is called **tidal locking**, and it is the normal outcome for a moon orbiting close to a much larger body. The hemisphere we never see from Earth is the **far side** — not the "dark side." It receives just as much sunlight as the near side.

Two kinds of terrain are visible with the naked eye. The number of craters on a surface tells you how old it is: the longer a surface has been there, the more impacts it has collected.

- **Highlands** — the bright, rugged regions. This is the Moon's **original crust**, so it is the oldest surface. It has been exposed to meteorite impacts for the whole history of the Moon, which is why it is so heavily cratered
- **Maria** (singular *mare*, Latin for "sea") — the dark, flat plains. Later, huge impacts punched basins into the crust, and lava welled up and flooded them. When that lava cooled it created a **new, smooth surface**, wiping out the craters that were there before. Because this

happened after the highlands formed, the maria have had less time to collect impacts — so they are only lightly cratered

••• You do • Check your understanding

Q1.8
Explain why we always see the same face of the Moon from Earth.

Q1.9
Describe the difference between the maria and the highlands, and state which is older.

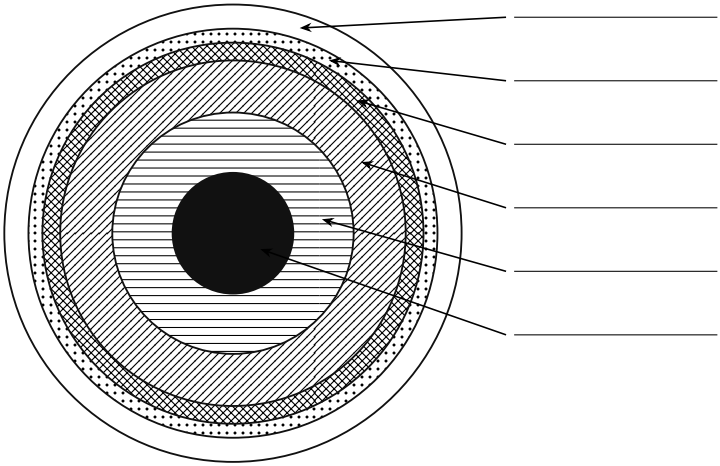
Q1.10
The Moon has no atmosphere. Explain how this causes its extreme temperature range.

1.4 Structure of the Sun

The Sun is a **main sequence star** — it converts hydrogen into helium through **nuclear fusion** in its core.

- **Core**: where fusion occurs; temperature ~15 million°C
- **Radiative zone**: energy moves outward as photons (takes ~100,000 years to cross)
- **Convective zone**: energy carried by convecting plasma
- **Photosphere**: the visible surface (~5,778 K); where sunspots form
- **Chromosphere and Corona**: outer atmosphere; much hotter than photosphere (corona: 1–3 million K)

Diagram: Label the layers of the Sun



Word bank: Convective zone • Corona • Core • Photosphere • Chromosphere • Radiative zone

• I do • Worked example: Light travel time from the Sun

The Sun is 150,000,000 km from Earth. Light travels at 300,000 km/s. How long does sunlight take to reach us?

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{150{,}000{,}000\ \text{km}}{300{,}000\ \text{km/s}} = 500\ \text{s} = 8\ \text{min}\ 20\ \text{s}$$

Answer: Sunlight takes 8 minutes and 20 seconds to reach Earth. This means the Sun we see right now is actually how it looked 8 minutes ago.

•• We do • Your turn

A flash of light leaves the Moon. How long before we see it? (Moon is 384,400 km away; speed of light = 300,000 km/s)

••• You do • Check your understanding

Q1.11
Define nuclear fusion as it occurs in the Sun.

Q1.12
State the temperature of the Sun's core and of its photosphere.

Q1.13
Jupiter is 5.20 AU from the Sun. Sunlight reaches Earth (1 AU) in 500 seconds. Calculate how long sunlight takes to reach Jupiter, in minutes and seconds.

▲ Above level • Year 8 extension

Year 8 Extension: How the Moon formed — the Giant Impact Hypothesis

The leading model for the Moon's origin is the **Giant Impact Hypothesis**. About **4.5 billion years ago** a Mars-sized body — named **Theia** — struck the young Earth. The collision threw a disc of molten rock and vapour into orbit, and that material clumped together under gravity to form the Moon.

Two pieces of evidence support it:

- The Moon has almost **no iron core**. Earth's iron had already sunk to its centre before the impact, so the debris blasted into orbit came mostly from the rocky mantle.
- Moon rock returned by the Apollo missions has **oxygen isotope ratios identical to Earth's**. Rock formed elsewhere in the solar system does not match — so the Moon and Earth are made of the same material.

Why it matters: the Moon's mass stabilises Earth's axial tilt at about 23.5°. Without it, the tilt could wander between 0° and 85° over millions of years, making stable long-term climates — and possibly complex life — far less likely.

▲ Above level • Year 8 extension

Year 8 Extension: Earth's magnetosphere and space weather

The liquid iron in Earth's **outer core** is constantly moving, and moving metal generates a magnetic field — the **dynamo effect**. That field does not stop at the surface. It reaches tens of thousands of kilometres out into space, forming the **magnetosphere**: an invisible shield around the whole planet.

It has something to shield us from.

- **Solar wind** — a constant stream of charged particles blowing outward from the Sun at 400–800 km/s. It never stops.

- **Coronal mass ejections** (CMEs) — huge bursts of plasma thrown off the Sun, which can reach Earth in one to three days.
- **Auroras** — the magnetosphere funnels charged particles toward the poles, where they collide with atmospheric gases and make them glow. This is the *aurora australis* in the south and the *aurora borealis* in the north. An aurora is what the shield looks like while it is working.

Space weather is the name for these events, and it has consequences on the ground. On 13 March 1989 a geomagnetic storm induced currents in Quebec's power transmission lines large enough to collapse the entire provincial grid in under 90 seconds, leaving about six million people without electricity for nine hours. Modern GPS, satellites and radio communication are all vulnerable.

Extension challenge. Mars has no global magnetic field, and most of its atmosphere has been stripped away by the solar wind over billions of years. Earth has both a field and an atmosphere. What does the comparison suggest the magnetic field is doing — and what might happen if Earth's field weakened a great deal?

Glossary

Every key word in this section and the page it is introduced on.

Word	Page	In my own words
Auroras	15	
Coronal mass ejections	15	
Crust	7	
Dynamo effect	14	
Empty space	5	
Giant Impact Hypothesis	14	
Highlands	9	
Inner core	7	
Magnetosphere	14	
Main sequence star	11	
Mantle	7	
Maria	9	
Nuclear fusion	11	
Outer core	7	
Regolith	9	
Satellite	9	
Solar wind	14	
Space weather	15	
Tidal locking	9	